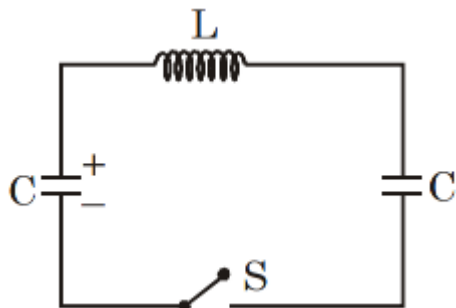


Q1

Figure shows an electric circuit with negligibly small active resistance. Initially left capacitor is charged to a potential V_0 and then the switch was closed.



- (1) Charge on right capacitor is given by $\frac{CV_0}{2}(1 - \cos \omega t)$; $\omega = \sqrt{\frac{2}{LC}}$
- (2) At the instant when charge on capacitor plates have same magnitude, total energy in capacitor and inductor is equal.
- (3) Maximum current in the circuit has magnitude $V_0 \sqrt{\frac{C}{2L}}$.
- (4) Maximum magnitude of induced emf in circuit is V_0 .

Q2

A process for monoatomic gas is defined by $V = aT^2$, where a is a constant. In this process temperature of 1 mole of a gas is increased by 100 K; then (V = Volume, T = Temperature)

- (1) Molar heat capacity of this process is negative.
- (2) The work done by the gas in the process is $250R$.
- (3) The work done by the gas in the process is $200R$.
- (4) The work done is given by $\int_{v_1}^{v_2} P dv$ in which P is a function of volume.

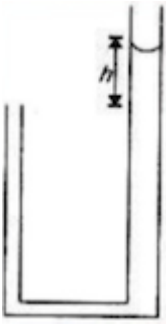
Q3

Two capillaries of small cross section are connected as shown in the figure. The right tube has cross sectional radius R and left one has a radius of $r (< R)$. The tube of radius R is very long where as the tube of radius r is of short length. Water is slowly poured in the right tube. Contact angle for the tube wall and water is $\theta = 0^\circ$. Let h be the height difference between water surface in the right and left tube. Surface tension of water is T

Questions with Answer Keys

MathonGo

and its density is ρ . Mark the CORRECT option(s)



- (1) The value of h if the water surface in the left tube is found to be flat is $h = \frac{2T}{R\rho g}$
- (2) The value of h if the water surface in the left tube is found to be flat is $h = \frac{T}{R\rho g}$
- (3) The maximum value of h for which water will not flow out of the left tube is $\frac{2T}{\rho g} \left(\frac{r+R}{rR} \right)$
- (4) The maximum value of h for which water will not flow out of the left tube is $\frac{T}{\rho g} \left(\frac{r+R}{rR} \right)$

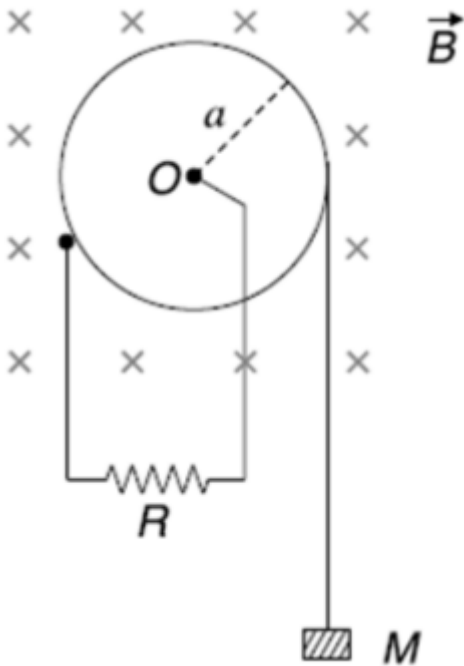
Q4

A metallic pulley is in the shape of a disc of radius a . It can rotate freely about a horizontal axis passing through its centre. The moment of inertia of the pulley about this axis is I . A light string is tightly wrapped around the pulley with its one end connected to a block of mass M . The centre of the pulley and its circumference are connected to a resistance R as shown. The contact of resistance at the circumference does not cause any friction. A uniform magnetic field B is switched on which is parallel to the axis of rotation of the pulley (see Figure). The mass M is allowed to fall. Assume that resistivity of the material of the pulley is negligible.

Questions with Answer Keys

MathonGo

The acceleration of the block of mass M at the instant its velocity becomes v_0 is



$$(1) \frac{Mga^2 - \frac{B^2 a^4 v_0}{4R}}{I + Ma^2}.$$

Assuming that the block can fall through a large distance, the terminal speed (v_T) that it will

$$(2) \text{acquire is } \frac{4MgR}{B^2 a^2}.$$

The rate of change of kinetic energy of the pulley at the instant when the speed of the falling

$$(3) \text{block is } \frac{v_T}{2} \text{ is equal to } \frac{IM^2 g^2 R}{B^2 a^2 (I + Ma^2)}.$$

The rate of change of kinetic energy of the pulley at the instant when the speed of the falling

$$(4) \text{block is } \frac{v_T}{2} \text{ is equal to } \frac{IM^2 g^2 a^2}{B^2 R (I + Ma^2)}.$$

Q5

A pulse is started at a time $t = 0$ along the $+x$ direction on a long, taut string. The shape of the pulse at $t = 0$ is given by function $f(x)$ with

$$f(x) = \begin{cases} \frac{x}{4} + 1 & \text{for } -4 < x \leq 0 \\ -x + 1 & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

here f and x are in centimeters. The linear mass density of the string is 50 g/m and it is under a tension of 5 N

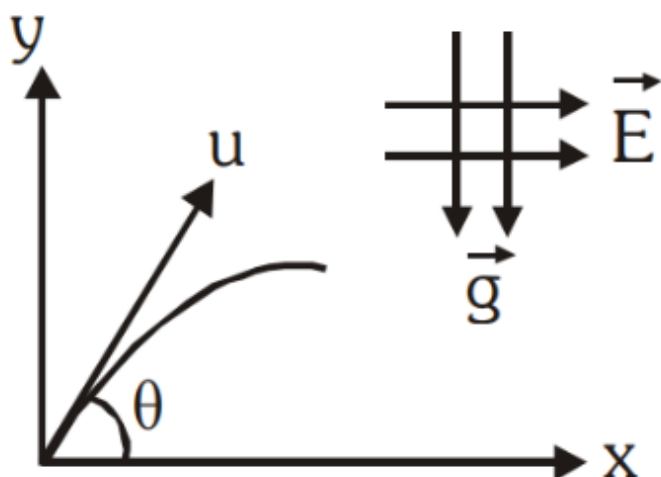
Questions with Answer Keys

MathonGo

- (1) The shape of the string is drawn at $t = 0$ then the area of the pulse enclosed by the string and the x -axis is 2.5 cm^2 .
- (2) The shape of the string is drawn at $t = 0$ then the area of the pulse enclosed by the string and the x -axis is 5 cm^2 .
- (3) The transverse velocity of the particle at $x = 13 \text{ cm}$ and $t = 0.015 \text{ s}$ will be -250 cm/s
- (4) The transverse velocity of the particle at $x = 13 \text{ cm}$ and $t = 0.015 \text{ s}$ will be 250 cm/s

Q6

A particle of mass m and charge $-q$ has been projected from ground as shown in the figure below.

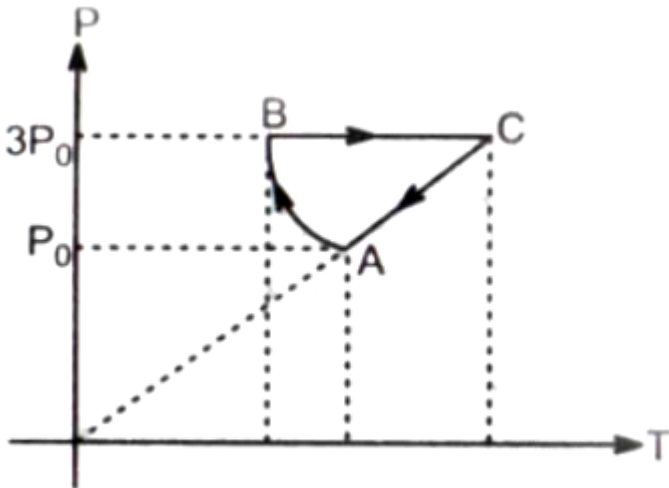


Mark out the correct statements (s) :-

- (1) The path of motion of the particle is parabolic
- (2) The path of motion of the particle is a straight line
- (3) Time of flight of particle is $\frac{2u \sin \theta}{g}$
- (4) Range of motion of the particle will be less than $\frac{u^2 \sin 2\theta}{g}$

Q7

One mole of monoatomic gas is taken through cyclic process shown below. $T_A = 300 \text{ K}$. Process AB is defined as $PT = \text{constant}$

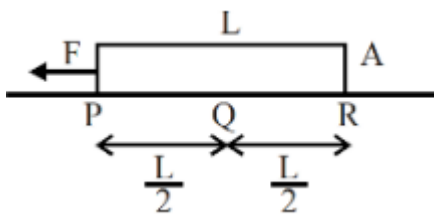


- (1) work done in process AB is $-400R$
- (2) change in internal energy in process CA is $900R$
- (3) heat transferred in the process BC is $2000R$
- (4) change in internal energy in process CA is $-900R$

Q8

A uniform rod of length L & cross section area ' A ' is kept on smooth surface. A force F is applied as shown.

Young's modulus = Y . Which of the following is/are INCORRECT :-



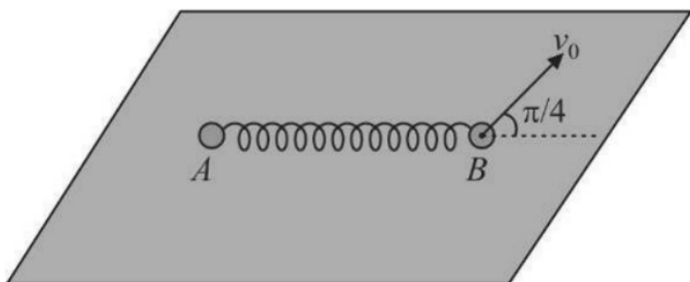
- (1) Elongation in the rod = $\frac{FL}{AY}$.
- (2) Stress at every cross section of rod = $\frac{F}{A}$.
- (3) Elongation in part PQ will be more than that in part QR .
- (4) Tension in the rod varies non-linearly with distance from P .

Q9

Questions with Answer Keys

MathonGo

A dumbbell consists of two balls A and B each of mass $m = 1 \text{ kg}$ and connected by a spring. The whole system is placed on a smooth horizontal surface as shown in the figure. Initially the spring is at its natural length, the ball B is imparted a velocity $v_0 = \frac{8}{\sqrt{7}} \text{ m/s}$ in the direction shown. The spring constant of the spring is adjusted so that the length of the spring at maximum elongation is twice that of the natural length of the spring. Find the maximum potential energy stored (in Joule) in the spring during the motion.



Q10

The work function of a cathode is estimated using light of wavelength $310 \pm 1 \text{ nm}$. The stopping potential is measured to be $500 \pm 1 \text{ mV}$. $\frac{hc}{e} = 1240 \text{ eV nm}$ (known precisely). What is the error e (in %) in work function. Find value of $10e$ by rounding off to nearest integer.

Q11

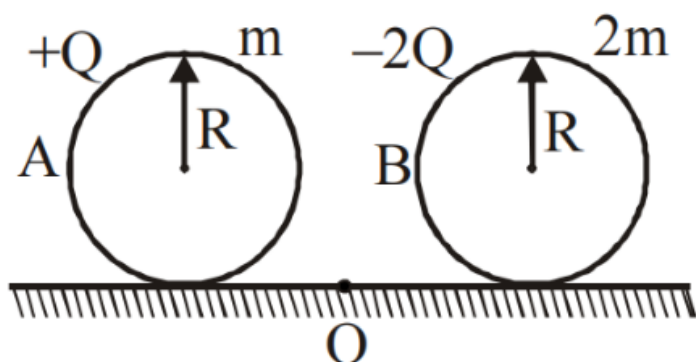
A nonconducting disc of radius R and uniform positive surface charge density σ is placed on the ground with its axis vertical. A particle of specific charge $\frac{q}{m} = \frac{4\epsilon_0 g}{\sigma}$ is dropped along the axis of the disc from a height h . The value of h if the particle just reaches the disc is $\left(2\sqrt{\frac{R}{3}}\right)^m$. Determine the value of m .

Q12

A hollow non conducting sphere A and a solid non conducting sphere B of equal radius 'R' and masses m and $2m$ are kept at a large distance apart on a rough horizontal surface. Charge on the two spheres A and B are Q and $-2Q$ respectively. Charges are distributed uniformly and remain constant and uniform as the spheres come closer. Friction is sufficient to support pure rolling and the kinetic energy of the two spheres just before collision is K_A and K_B .

Questions with Answer Keys

MathonGo

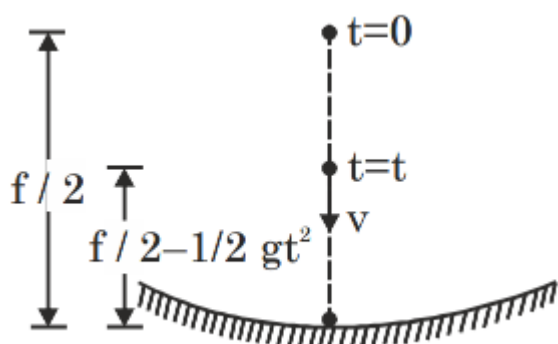
Find $100 \times \frac{K_A}{K_B}$ 

Q13

A transparent lift A is going upwards with velocity 20 ms^{-1} and retarding at the rate of 8 ms^{-2} . Second transparent lift B is located in front of it and is going down at 10 ms^{-1} with retardation of 2 ms^{-2} . At the same instant a bolt from the ceiling of lift A drops inside lift A. If height of car of lift A is 16 m then find the distance travelled by bolt as observed by a person in lift B till the time it collides with floor of lift A.

Q14

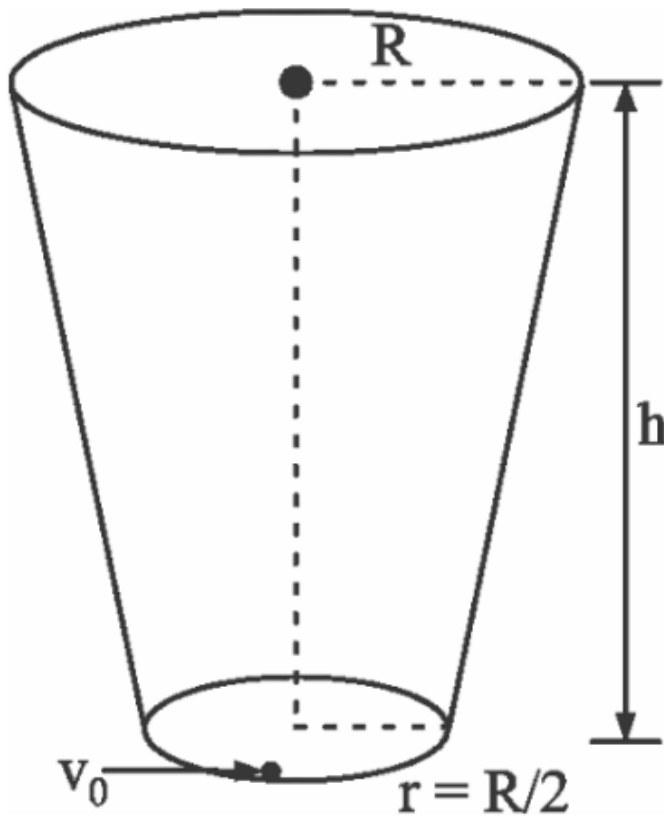
A particle is dropped along the axis from a height $f/2$ on a concave mirror of focal length f as shown in figure. The acceleration due to gravity is g . The maximum speed of image is $\frac{3}{4}\sqrt{nfg}$. Find n



Q15

A small particle of mass m lies on the bottom surface of smooth conical glass surface, as shown in the figure. The particle touches the circumference. The minimum velocity v_0 (along the circumference) such that particle

reaches at top of surface is :



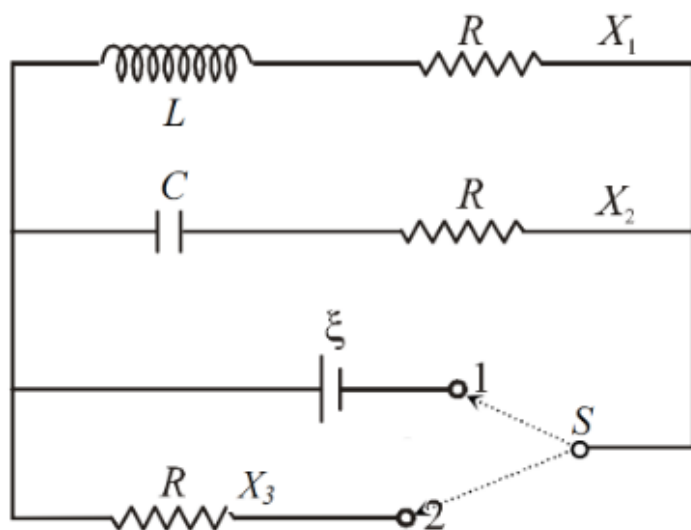
- (1) $\sqrt{\frac{8}{3}gh}$
- (2) $2\sqrt{gh}$
- (3) $\sqrt{\frac{2}{3}gh}$
- (4) $\sqrt{gh}$

Q16

Figure shows a circuit containing three resistor X_1 , X_2 & X_3 having resistance R each, an inductor, capacitor and an emf source having inductance 'L', capacitance C & emf ξ respectively (Given $R = \sqrt{\frac{L}{C}}$). The switch is first connected to position-1. When charge on capacitor becomes half of its maximum possible value then switch S is connected to position-2. Current in resistance X_3 just after shifting the switch from position-1 to position-2, is

Questions with Answer Keys

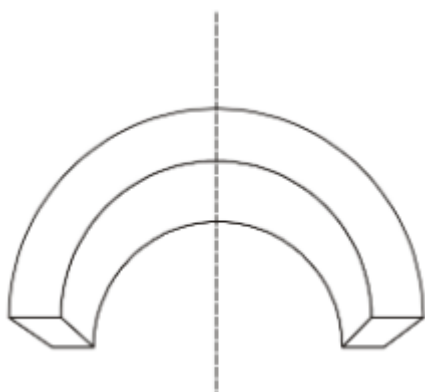
MathonGo

(given $L = 5\text{mH}$, $C = 2\mu\text{F}$, $R = 10\Omega$, $\xi = 5\text{ V}$)

- (1) 1
 (2) 1.5
 (3) 2.5
 (4) None of these

Q17

A conducting rod of square cross section is bent in a semi circular shape of radius ' r ' as shown in figure. Length of side of cross section is equal to ' a '. Resistivity of material is ' ρ '. The resistance of rod across its end will be



- (1) $\frac{\pi\rho}{a \ln(1+a/r)}$
 (2) $\frac{2\pi\rho}{a \ln(1+a/r)}$

Questions with Answer Keys

MathonGo

(3) $\frac{\pi\rho(r+a)}{a^2}$

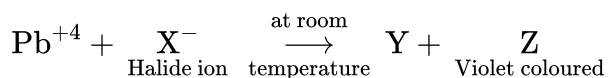
(4) $\frac{2\pi r\rho}{a^2}$

Q18

A radioactive sample contains two different types of radioactive nuclei: A-with half-life 5.0 days and B-with half-life 30.0 days. Initially the decay rate of the A type nuclei is 64 times that of B type nuclei. Their decay rates will be equal when time is

- (1) 9 days
- (2) 18 days
- (3) 27 days
- (4) 36 days

Q19

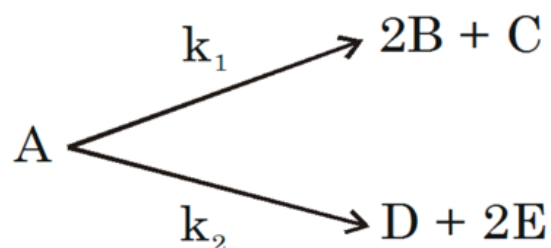


Statements which is/are correct will be :-

- (1) Oxidation state of Pb in 'Y' is +2
- (2) X^- act as reducing agent in this reaction
- (3) Gas 'z' is diatomic
- (4) Given reaction is redox reaction

Q20

If an optically active compound 'A' decompose through given parallel 1st order kinetics.



Initial mole of A is 2 ($k_1 = 0.0693 \text{ sec}^{-1}$, $k_2 = 0.1386 \text{ sec}^{-1}$). If only A, B & D are optically active

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Questions with Answer Keys

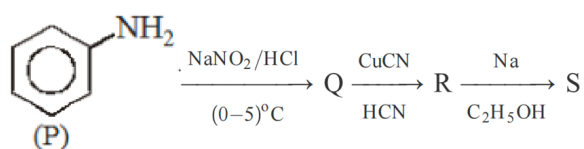
MathonGo

compounds and their angle of rotation per mole are 60° , 30° , -90° respectively, then which of the following is/are correct -

- (1) Total angle of rotation of mixture after time $(10/3)$ sec is 20°
- (2) Mixture is levorotatory after 75% decomposition of A
- (3) Overall $t_{1/2}$ of A is $10/3$ sec.
- (4) Percentage of C formed in the product is 11.11%

Q21

Correct statement(s) regarding following reaction sequence is/are :



- (1) P & S can be distinguished by dye azo test
- (2) S is more basic than P
- (3) R give benzophenone with PhMgBr followed by H_3O^+
- (4) Q on passing with steam produces phenol

Q22

A solution contains Na_2CO_3 and NaHCO_3 . 10ml of this solution requires 2.5ml of $\frac{1}{5}\text{M}$ HCl solution for the end point using phenolphthalein as indicator. In another experiment 10ml of the same original solution requires 7.5ml of $\frac{1}{5}\text{M}$ HCl solution for the end point using methyl orange as indicator. Which of the following statement(s) is/are correct regarding the original solution. [Atomic mass : Na = 23]

- (1) 10ml of original solution contains 0.053 g Na_2CO_3
- (2) 10ml of original solution contains 0.042 g NaHCO_3 .
- (3) Concentration of Na_2CO_3 in original solution 0.05M
- (4) Concentration of NaHCO_3 in original solution 0.05M

Q23

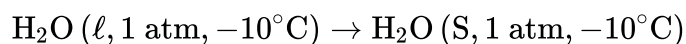
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Questions with Answer Keys

MathonGo

Irreversible process is :



$$\text{Given at } 0^\circ\text{C} \quad \Delta H_{\text{fusion}} = 6 \text{ kJ/mole}$$

$$\text{at } -10^\circ\text{C} \quad \Delta H_{\text{fusion}} = 5.55 \text{ kJ/mole}$$

$$C_p(\text{ice}) = 30 \text{ J/K-mole}$$

$$C_p(\text{water}) = 75 \text{ J/K-mole}$$

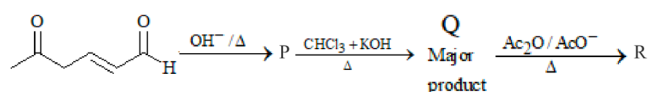
$$\ln \frac{273}{263} = 0.04$$

Pick out the incorrect statements :

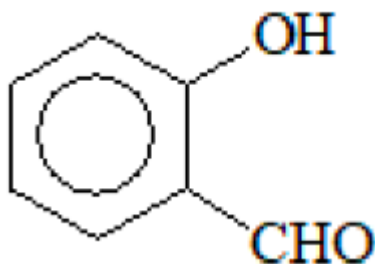
- (1) ΔS_{total} for process = -20.4 J/K-mole
- (2) ΔS_{surr} for process = 21.1 J/K-mole
- (3) ΔS_{total} for process = 0.92 J/K-mole
- (4) ΔS_{total} for process = 22 J/K-mole

Q24

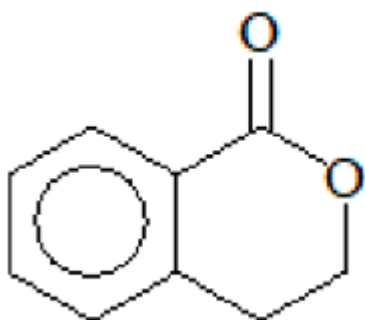
Observed the following sequence of reaction :



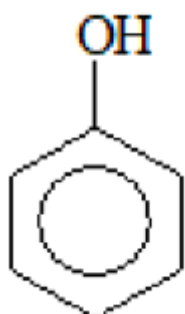
Which of the following is/are correct?



- (1) Product Q is



(2) Product R is



(3) Product P is

(4) Reimer Tiemann reaction is involved

Q25

A lead storage cell is discharged which causes the H_2SO_4 electrolyte to change from a concentration of 40% by weight (density = 1.260 gm/ml) to 28%, by weight. The original volume of electrolyte was one litre.

Identify the correct statement(s) :-

- (1) The overall cell reaction is: $\text{Pb(s)} + \text{PbO}_2(\text{s}) + 2\text{H}_2\text{SO}_4(\text{aq}) \rightarrow 2\text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\ell)$.
- (2) A total of 2.0 moles of H_2SO_4 have reacted.
- (3) The total charge released from anode of the cell is 1.93×10^5 coulomb.
- (4) The mass of electrolytic solution has decreased.

Q26

Choose the correct statement(s) among the following :

- (1) $[\text{FeCl}_4]^-$ has tetrahedral geometry.
- (2) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$ has 2 geometrical isomers.

Questions with Answer Keys

MathonGo

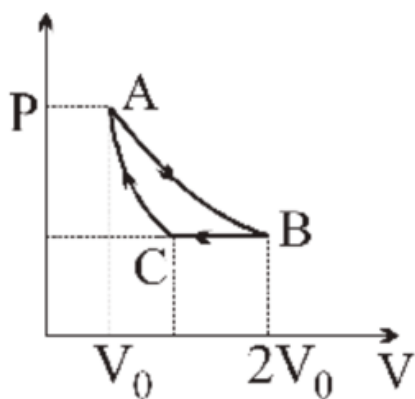
(3) $[\text{FeCl}_4]^-$ has higher spin-only magnetic moment than $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$.

(4) The cobalt ion in $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$ has $\text{sp}^3 \text{d}^2$ hybridization.

Q27

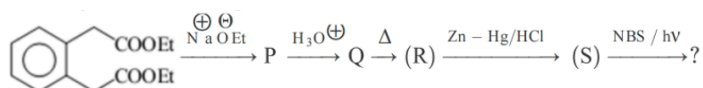
In a cycle ABCA consisting of isothermal expansion AB, isobaric compression BC & adiabatic compression CA. Find the % efficiency of cycle. (Give your answer as nearest integer)

[Given : $T_A = T_B = 400 \text{ K}$; $\gamma = 1.5$, $\ln 2 = 0.693$, $2^{\frac{1}{3}} = 1.26$]



Q28

For the given reaction sequence :



Number of monohalogenated products is/are -

Q29

An undergraduate student heated 1-cyclopentylethyl ethanoate at 300°C to obtain an unsaturated hydrocarbon 'K' which was then treated with mercuric acetate followed by water then sodium borohydride to give 'L' which is dehydrated by conc. H_2SO_4 at 180°C to give 'M' as one of products. What is the molecular mass of compound 'M'?

Q30

Questions with Answer Keys

MathonGo

How many of the following salts are white and evolves colourless gas or vapours with dil. H_2SO_4 .

BaCrO_4 , Hg_2CrO_4 , ZnS , BaSO_4 , BaS_2O_3 , CH_3COOAg , AgNO_2 .

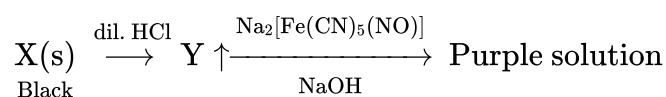
Q31

Gaseous nitrosyl chloride and nitrogen are taken in a flask sealed and heated to some temperature where the total pressure would have been 1 bar if the following equilibrium had not been established.



But the actual pressure was found to be 1.2 bar due to above equilibrium. Now into the above equilibrium mixture, some Cl_2 gas was introduced at constant V & T so that the total pressure would have been 8.3 bar if no further reaction had occurred but the actual pressure was found to be 8.2 bar. find the value of K_P .

Q32



Gas Y has been allowed to react with following species in neutral/acidic medium :

- (a) FeCl_3
- (b) CuSO_4
- (c) BaCl_2
- (d) SO_2
- (e) $\text{Cr}_2\text{O}_7^{2-}$
- (f) CH_3COONa
- (g) Hg^{2+}

Then calculate value of $(P - Q)/R$:

P : Number of species which undergoes redox reaction with gas Y.

Q : Number of species with which gas Y undergoes precipitation without going into redox reaction.

R : Number of species with gas Y produce no observable change.

Q33

Questions with Answer Keys

MathonGo

Dissolution of solids in water can be exothermic or endothermic process but gases dissolve in water always with the evolution of heat. Dissolution of a substance in water can be either because ion dipole interactions or by hydrogen bond formation. Pressure plays a significant role on the solubility to gases in water. Solubility of a gas in terms of mol fraction (X_B) is related to pressure (P) according to the mathematical relation

$$P = K_H X_B$$

On the basis of above paragraph answer the following questions.

When a pinch of salt (NaCl) is added to a freshly opened bottle of Coca-Cola or Limca, a lot of effervescence occurs with evolution of a colourless gas

- (1) The gas evolved is O_2
- (2) The gas evolved is HCl
- (3) There is no evolution of gas only decomposition of NaCl occurs
- (4) CO_2 is evolved due to its displacement from aerated soft drink

Q34

Which of the following is/are correct?

- (1) Langmuir adsorption theory states that reactivity of a site also depends on the vacancy status of nearby sites.
- (2) Order of adsorption can be changed by changing pressure conditions.
- (3) Desorption of gas on solid is favoured by low temperature.
- (4) ZSM-5 is a shape independent catalyst used to crack hydrocarbons.

Q35

Consider the following order.

I. Thermal stability: $\text{BeSO}_4 < \text{MgSO}_4 < \text{CaSO}_4 < \text{SrSO}_4 < \text{BaSO}_4$

II. Bond angle: $\text{OCl}_2 > \text{SF}_2 > \text{AsH}_3 > \text{H}_2\text{Se}$

III. Bond order of O_2^+ , NO and N_2^- is 2.5

Questions with Answer Keys

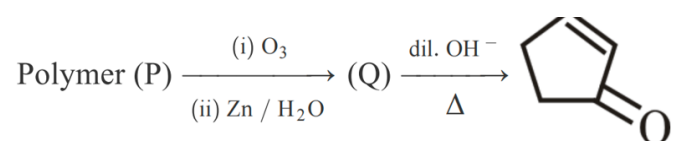
MathonGo

IV. NCl_3 on hydrolysis gives HOCl but BCl_3 , gives HCl .

Identify which of the above four statements are correct.

- (1) I, II, III and IV
- (2) II, III and IV
- (3) I, II and IV
- (4) I and II

Q36



Incorrect for above reaction sequence is :

- (1) (P) exist in both cis and trans forms
- (2) (P) is a 1,4-addition polymer of 2-methylbuta-1,3-diene
- (3) Q can show both Tollen's and Iodoform test.
- (4) Q gives positive Iodoform test but no reaction with NaHSO_3

Q37

If $f(x) = 2 + \int_{-1}^1 \left(\frac{tx^2}{2} + \frac{9x}{14} \right) f(t) dt$, then:

- (1) Rolle's Theorem is applicable for $y = f(x)$ in $[-2, -1]$
- (2) $\lim_{x \rightarrow 0} f(x) = 0$
- (3) f is continuous and derivable on R
- (4) maximum value of $f(x)$ does not exist

Q38

Let $E - ABCD$ be a pyramid on square base $ABCD$ where A is the origin and B and D are lying on positive x -axis and y -axis respectively. If E is $(0, 2, 3)$ and $\overrightarrow{DE} \cdot (\hat{i} + \hat{j}) = \vec{0}$, then:

Questions with Answer Keys

MathonGo

- (1) image of the point D in the plane ABE is $\left(0, \frac{-10}{13}, \frac{24}{13}\right)$
- (2) image of the point D in the plane ABE is $\left(0, \frac{-6}{13}, \frac{30}{13}\right)$
- (3) volume of the tetrahedron $ABDE$ is 2 cubic units
- (4) perpendicular distance of the point D from the plane ABE is $\frac{9}{\sqrt{13}}$

Q39

A bag contains 3 red and 3 green balls and a person draws out 3 at random. He then drops 3 blue balls into the bag and again draws out 3 at random. The probability that the 3 later balls are all of different colours is equal to -

- (1) $\frac{27}{50}$
- (2) $\frac{27}{100}$
- (3) $20 \left(\frac{{}^3C_1 \cdot {}^2C_1}{{}^6C_3} \right)^3$
- (4) probability of x satisfying the inequality $x^2 - 33x + 90 \leq 0$ such that $x \in [1, 100]$

Q40

Let two parabolas be $S_1 : y^2 = 4ax$ and $S_2 : y^2 = -4ax$. From any point P on S_1 , tangents are drawn to S_2 touching it at Q and R , then:

- (1) line QR is tangent to S_1
- (2) line QR neither touches nor intersect S_1
- (3) if normal at any point $A(t)$ on S_1 is tangent to S_2 then $t^2 = \sqrt{2} - 1$
- (4) if normal at any point $A(t)$ on S_1 is tangent to S_2 then $t^2 = \sqrt{2} + 1$

Q41

Consider a cube of side length 9 units. Let (x, y, z) be coordinates of points on or inside the cube such that $x, y, z \in \mathbb{I}$ and $0 \leq x, y, z \leq 9$. If total number of ways of selecting two distinct points among these such that their mid-point is also having integral coordinates is N , then

- (1) N is divisible by 30

Questions with Answer Keys

MathonGo

- (2) N is divisible by 31
- (3) N is divisible by 32
- (4) Number of factors of N is 40

Q42

The values of 't' which satisfies $(t - [\sin x])! = 3!5!7!$ is/are (where $[.]$ denotes greatest integer function) -

- (1) 9
- (2) 10
- (3) 11
- (4) 12

Q43

Which of the following statement(s) are correct -

- (1) If $f(x) \equiv 6x^7 + 5x^6 + 4x^5 + 3x^4 + 2x^3 + x^2 + x + 1$, then the equation $f'(x) = 0$ must have a real root.
- (2) If $(x - 2)$ is a factor of the polynomial $P(x)$ (degrees 5) repeated 3 times, then 2 is the root of the equation $P'(x) = 0$ repeated 2 times
- (3) If $f(x)$ is a differentiable function, then if its graph is symmetric about origin, then the graph of $f'(x)$ will be symmetric about y-axis.
- (4) If $y = \sin^{-1}(\cos(\sin^{-1} x)) + \cos^{-1}(\sin(\cos^{-1} x))$ then $\frac{dy}{dx}$ is independent of x .

Q44

Let $f : R \rightarrow (0, 1)$ be a continuous function, then which of the following pair of vectors are linearly dependent for some $x \in (0, 1)$?

- (1) $\vec{a} = f(x)\hat{i} + 2\hat{j}; \vec{b} = x^2\hat{i} + 3\hat{j}$
- (2) $\vec{a} = f(x)\hat{i} + 3\hat{j}; \vec{b} = x^2\hat{i} + 2\hat{j}$
- (3) $\vec{a} = \left(\int_0^{1-x} f(t)dt\right)\hat{i} + 3\hat{j}; \vec{b} = x\hat{i} + 2\hat{j}$
- (4) $\vec{a} = \left(\int_0^{1-x} f(t)dt\right)\hat{i} + 2\hat{j}; \vec{b} = x\hat{i} + 3\hat{j}$

Q45

Let λ is the coefficient of x^2 in the expansion $(1+x)(1-3x)(1+5x)(1-7x)\dots(1-23x)(1+25x)$ then number of positive divisors of $|\lambda|$ is

Q46

Two numbers are selected independently at random in the interval $[0, 1]$. If the smaller one is less than $1/3$, then find the probability that the larger one is greater than $3/4$.

Q47

On the ellipse $E = \frac{x^2}{64} + \frac{y^2}{9} = 1$, tangents drawn at the point $P_1, P_2, P_3, \dots, P_n$ on the ellipse intersecting the major axis at $T_1, T_2, T_3, \dots, T_n$ respectively. If the value of $\sum_{i=1}^n \frac{\text{Area}(\Delta P_i T_i S) \cdot \text{Area}(\Delta P_i T_i S')}{(P_i T_i)^2} = 18$, where S and S' represents the foci of the ellipse, then 'n' equal to:

Q48

Let $S_1 \equiv x^2 + y^2 - 4x - 8y + 4 = 0$ and S_2 its image in the line $y = x$. The radius of the circle touching $y = x$ at $(1, 1)$ and orthogonal to S_2 is $\frac{3}{\sqrt{\lambda}}$, then $\frac{\lambda^3}{5}$ is equal to

Q49

Given two curves : $y = f(x)$ passing through $(0, 1)$ and $y = \int_{-\infty}^x f(t)dt$ passing through $(0, \frac{1}{3})$. The tangents drawn to both the curves at the points with equal abscissas intersects on x -axis. Find the value of $\ln f(3)$

Q50

If $\begin{bmatrix} 1 & 0 & 0 \\ 6 & 2 & 0 \\ 5 & 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ x^2 \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ 2ax + bx^2 \\ 5x + cx^2 + 3 \end{bmatrix} \forall x \in \mathbb{R}$ and $f(x)$ is a differentiable function satisfying $f(x) + f(y) = f\left(\frac{x+y}{1-xy}\right)$ for all $x, y \in \mathbb{R}, (xy \neq 1)$ and $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 2$, then find the value of $\left[\int_0^1 \frac{ax^2+bx+c}{f(1)} dx \right]$, where $[.]$ denotes greatest integer function.

Q51

For $2n$ observations $a_1, -a_1, a_2, -a_2, a_3, -a_3, \dots, a_n, -a_n$ where all a_i 's are distinct. Let $\bar{x}, \sigma$ and M denotes mean, standard deviation and median, respectively. Then which of the following option is incorrect when n is an odd integer?

- (1) $\bar{x} = 0$
- (2) $\bar{x} = M$
- (3) $\sigma > \bar{x}$
- (4) $\sigma < M$

Q52

A is the $n \times n$ matrix whose elements are all '1' and B is the $n \times n$ matrix whose diagonal elements are all ' n ' and other elements are ' $n - r$ '. Then, A^2 is a scalar multiple of A and then $(B - rI) [B - (n^2 - nr + r) I]$ is

- (1) I
- (2) -I
- (3) 0
- (4) None of these

Q53

Find the number of all 5 -digit numbers each of which contains the block 15 and is divisible by 15 ? (Example of such number is 31545, 34515 etc)

- (1) 479
- (2) 487
- (3) 480
- (4) None of these

Q54

Questions with Answer Keys

MathonGo

If sum of the roots of the equation

$$\sqrt{2} \sin\left(x + \frac{\pi}{4}\right) \left(\sec^2 x - (1 + \sqrt{3}) \tan x + 2 + \sqrt{3}\right) = \sqrt{9 + 18 \sin x \cdot \cos x} \text{ in } [0, 2\pi] \text{ is } \frac{p}{q} \pi$$

where $p, q \in N$, then find the least value of $|p - q|$.

- (1) 25
- (2) 26
- (3) 30
- (4) 32

Answer Key

Q1 (1) (2) (3) (4)	Q2 (1) (3) (4)	Q3 (1) (3)	Q4 (1) (2) (3)
Q5 (1) (3)	Q6 (1) (3) (4)	Q7 (1) (3) (4)	Q8 (1) (2) (4)
Q9 (2)	Q10 (4)	Q11 (2)	Q12 (168)
Q13 (51)	Q14 (3)	Q15 (1)	Q16 (4)
Q17 (1)	Q18 (4)	Q19 (1) (2) (3) (4)	Q20 (1) (2) (3) (4)
Q21 (1) (2) (3) (4)	Q22 (1) (2) (3) (4)	Q23 (1) (4)	Q24 (1) (3) (4)
Q25 (1) (2) (3) (4)	Q26 (1) (3)	Q27 (11)	Q28 (2)
Q29 (96)	Q30 (3)	Q31 (3.2)	Q32 (0.5)
Q33 (4)	Q34 (2)	Q35 (1)	Q36 (4)
Q37 (1) (2) (3) (4)	Q38 (1) (3)	Q39 (1) (3)	Q40 (1) (3)
Q41 (2) (4)	Q42 (2) (3)	Q43 (2) (3) (4)	Q44 (2) (3) (4)
Q45 (8)	Q46 (0.3)	Q47 (8)	Q48 (1.6)
Q49 (9)	Q50 (3)	Q51 (4)	Q52 (3)
Q53 (1)	Q54 (1)		